

Session 7

Aim

To study preemptive scheduling, Round Robin, priority scheduling, context switching, race conditions and deadlock.

```
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 12
On-Line CPU(s) list: 0-11
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-1334U
CPU family: 6
Model: 186
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 1
BogoMIPS: 4992.82
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpebgb rdts
cp lm constant_tsc rep_good nopt stopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 xlapic movbe popcnt tsc_deadline_timer avx512f avx512vbq avx512vnni avx512vbq avx512vbq avx512vbq avx512vbq avx512vbq
bpb stibp shrsr enhanced tpr shadow ept vpid ept-ad fsgsbase tsc_adjust hwp1 hwp2 hwp3 erms invpcid rdpseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsavei xsavev vmmi umip waitpkg gfni vaes vpclmulqdq rdp10 movdiri movdir64b fsrm md_
clear seriallize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: Full
Caches (sum of all):
L1d: 280 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-11
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
```

```
top - 08:06:18 up 1 min, 1 user, load average: 0.16, 0.09, 0.03
tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.7 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7754.6 total, 6279.1 free, 543.0 used, 1883.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7211.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S    %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 21892 13294  9720  S    0.0   0.2   0:01.62 systemd
    2 root        20   0  1180  2294  2072  S    0.0   0.0   0:00.02 init-systemd[ub
    6 root        20   0  3196  2128  2888  S    0.0   0.0   0:00.01 init
   58 root       19  -1 42168 16172 15188  S    0.0   0.2   0:00.35 systemd-journal
  160 root       20   0 25284 6584  5124  S    0.0   0.1   0:00.38 systemd-udev
  123 systemd+  20   0 11344 13184 11856  S    0.0   0.2   0:00.22 systemd-resolve
  127 systemd+  20   0  91836 7936  6972  S    0.0   0.1   0:00.15 systemd-timesyn
  187 root       20   0  4248  2684  2432  S    0.0   0.0   0:00.01 cron
  188 message+  20   0  9840  5356  4656  S    0.0   0.1   0:00.13 dbus-daemon
  198 root       20   0  39988  8888  7812  S    0.0   0.1   0:00.20 systemd-logind
  200 root       20   0 1756628 13864 11844  S    0.0   0.2   0:00.20 ml-pre-service
  218 root       20   0  3124  1960  1832  S    0.0   0.0   0:00.00 Agetty
  220 syslog    20   0 222516 5984  4588  S    0.0   0.1   0:00.15 rsyslogd
  277 root       20   0 107036 23144 13684  S    0.0   0.3   0:00.32 unattended-upgr
  327 root       20   0  3184  1188  980  S    0.0   0.0   0:00.08 SessionLeader
  328 root       20   0  3288  1248  1188  S    0.0   0.0   0:00.02 Relay(129)
  329 teja       20   0  6076  5388  3660  S    0.0   0.1   0:00.07 bash
  330 root       20   0  6696  4652  3876  S    0.0   0.1   0:00.03 login
  379 teja       20   0 20332 11488 9356  S    0.0   0.1   0:00.28 systemd
  380 teja       20   0  2160  2572  1848  S    0.0   0.0   0:00.00 (sd-pam)
  404 teja       20   0  6888  5360  3760  S    0.0   0.1   0:00.05 bash
  432 root       20   0 25288 3728 2228  S    0.0   0.0   0:00.00 (udev-worker)
  433 root       20   0 25288 3728 2228  S    0.0   0.0   0:00.00 (udev-worker)
  434 teja       20   0  9388  5416  3236  R    0.0   0.1   0:00.01 top
```

```

bhavan@LAPTOP-FGDPB110:~$ sleep 100 &
[1] 459
bhavan@LAPTOP-FGDPB110:~$ sleep 200 &
[2] 460
bhavan@LAPTOP-FGDPB110:~$ fg %1
sleep 100
bg %1
^C
bhavan@LAPTOP-FGDPB110:~$ bg %1
-bash: bg: %1: no such job
bhavan@LAPTOP-FGDPB110:~$ kill %1
-bash: kill: %1: no such job
bhavan@LAPTOP-FGDPB110:~$ kill %1
-bash: kill: %1: no such job
bhavan@LAPTOP-FGDPB110:~$ kill %2
bhavan@LAPTOP-FGDPB110:~$ pgrep sleep
[2]+  Terminated                  sleep 200
bhavan@LAPTOP-FGDPB110:~$ pgrep sleep
bhavan@LAPTOP-FGDPB110:~$ renice 5 -p PID
renice: bad process ID value: PID
bhavan@LAPTOP-FGDPB110:~$ ps -o pid,comm,pri,ni -p PID
error: process ID list syntax error

```

Usage:

ps [options]

Try 'ps --help <simple|list|output|threads|misc|all>'
 or 'ps --help <s|l|o|t|m|a>'
 for additional help text.

For more details see ps(1).

```

bhavan@LAPTOP-FGDPB110:~$ |

```

```

GNU nano 2.2.1                                context.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();
    if (pid == 0)
    {
        for (int i = 1; i <= 5; i++)
        {
            printf("Child process: %d\n", i);
            sleep(1);
        }
    }
    else
    {
        for (int i = 1; i <= 5; i++)
        {
            printf("Parent process: %d\n", i);
            sleep(1);
        }
    }
    return 0;
}

```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was

```

For more details see ps(1).
bhavan@LAPTOP-FGDPB110:~$ nano context.c
bhavan@LAPTOP-FGDPB110:~$ gcc context.c -o context
bhavan@LAPTOP-FGDPB110:~$ ./context
Parent process: 1
Child process: 1
Child process: 2
Parent process: 2
Child process: 3
Parent process: 3
Parent process: 4
Child process: 4
Parent process: 5
Child process: 5
bhavan@LAPTOP-FGDPB110:~$ |

```

Short theory

The parent and child processes execute alternately. The operating system saves one process state and loads another process state. This operation is called context switching.

```

bhavan@LAPTOP-FGDPB110:~$ vmstat 1 5
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r  b    swpd    free    buff    cache    si    so    bi    bo    in    cs  us  sy  id  wa  st  gu
 1  0      0 7356360    5212 196640      0      0   800    17  261      0  0  0 100  0  0  0
 0  0      0 7356360    5212 196640      0      0      0      0  176   68  0  0 100  0  0  0
 0  0      0 7356360    5212 196712      0      0      0      0  159   64  0  0 100  0  0  0
 0  0      0 7356360    5212 196712      0      0      0      0   73   36  0  0 100  0  0  0
 0  0      0 7356360    5212 196712      0      0      0      0   77   36  0  0 100  0  0  0
bhavan@LAPTOP-FGDPB110:~$ chrt -p $$
pid 329's current scheduling policy: SCHED_OTHER
pid 329's current scheduling priority: 0
bhavan@LAPTOP-FGDPB110:~$ |

```

in this session, we studied preemptive scheduling and context switching. Round Robin gives each process a fixed time quantum. Priority scheduling selects the process with the highest priority. We also studied race conditions, critical sections, deadlock, starvation and aging. A deadlock occurs when processes wait permanently for resources held by one another. Aging prevents starvation by gradually increasing the priority of a waiting process.